

Crop Price Forecast API Documentation

This documentation provides instructions on setting up and using the Crop Price Forecast API.

Setup

You can set up the API using Docker in two ways:

Option 1: Build and Run Locally

1. Ensure you have Docker installed.
2. Navigate to the root directory of the codebase.
3. Build the Docker image using the provided `Dockerfile`:

```
docker build -t crop-price-api .
```

4. Run the Docker container, mapping port 8080:

```
docker run -p 8080:8080 crop-price-api
```

5. The API should now be running locally. You can access the OpenAPI documentation at <http://localhost:8080/docs>.

Option 2: Pull from Docker Hub

1. Ensure you have Docker installed.
2. Pull the latest image from Docker Hub:

```
docker pull sllakshancc/crop-price-api:latest
```

3. Run the Docker container, mapping port 8080:

```
docker run -p 8080:8080 sllakshancc/crop-price-api:latest
```

4. The API should now be running locally. You can access the OpenAPI documentation at <http://localhost:8080/docs>.

Hosted App

Alternatively, you can use the hosted version of the app at: <https://data-crunch-app-1086781479342.us-central1.run.app/docs> (<https://data-crunch-app-1086781479342.us-central1.run.app/docs>)

How to Use the API

Important Note: Before using any prediction or evaluation endpoints, you **must** first invoke the `/api/train_scratch` endpoint to train and initialize the models from scratch.

Endpoints

Here's a description of the available endpoints:

- **POST /api/train_scratch**
 - Initiates the training of all models from scratch using historical data. This must be called before using other endpoints for the first time.
- **POST /api/data/weather**
 - Ingests new weather data records.
 - Triggers incremental training of models based on the new weather data.
 - Evaluates models and performs a full retraining from scratch if the NRMSE exceeds a threshold (1.55).
- **GET /api/data/weather**
 - Retrieves the latest weather data records for a specified region.
- **POST /api/data/prices**
 - Ingests new price data records.
 - Triggers incremental training for the models related to the commodities present in the new data.
 - Evaluates the models for the affected commodities and performs a full retraining from scratch if any of their NRMSE exceeds a threshold (1.55).
- **GET /api/data/price**
 - Retrieves the latest price data records for a specified commodity and region.
- **POST /api/predict**
 - Provides price predictions for a specified crop and region for the next 4 weeks.
- **POST /api/evaluate**
 - Evaluates all models and returns the overall RMSE and NRMSE, as well as individual evaluations per commodity.
- **POST /api/evaluate_model**
 - Evaluates the model for a specific commodity and returns its RMSE and NRMSE.

Data Ingress and Prediction Approach

The application employs a hybrid approach to data ingress and prediction, balancing the trade-off between latency and accuracy. When new data (weather or price) is ingested, the relevant models are updated through **online learning** (incremental training). After the incremental training, the models are evaluated. If the prediction error (specifically, the

Normalized Root Mean Squared Error - NRMSE) for new data is higher than a defined threshold (1.55), the system triggers a full **training from scratch** process using all available historical and newly ingested data. This ensures that the models are quickly updated with new data for most cases, while a full retraining occurs when necessary to maintain accuracy when the model's performance significantly degrades on new data.